



Instructor: Date:

Learner: Regional Branch:

Program: Center:

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Let's Start!

How should we figure out software development problems?



Learning Outcome:

Describe ways of solving problems and getting things done in written and spoken situations and in work and social contexts.

Train the system users on application structuring and management, according to the established plan and user profile, according to the organization policies.

Ask your partner:

- How could you make sure that your organization is being efficient when developing software?
- What quality and efficiency parameters do you think a company needs to follow for developing software?

Can Do Statements:

- I can relate the concept of performance metrics to the idea of measuring software quality.
- I can list some metrics used by software development organizations.
- I can discuss the need of evolving for metrics to measure software quality.



Let's Explore!

Developing software can be an arduous* job.

* We use the word *arduous* to refer to something that is long and hard to accomplish.

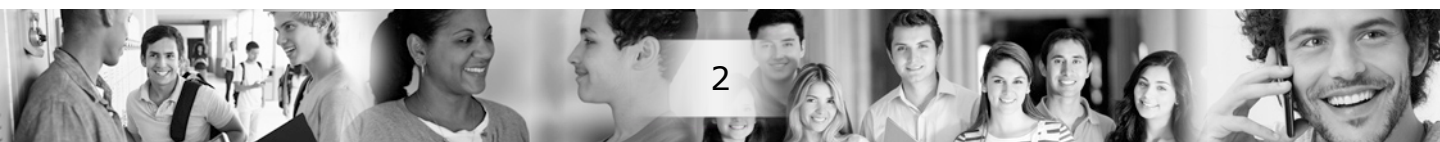


Instructions: Read the text below.

Developing software can be an arduous work as it needs a lot of planning, problems need to be fixed, and customers' requirements need to be fulfilled. That's why performance metrics need to be used by software organizations for easily spotting problems in the development processes throughout the development of a product. These metrics will help the manager and the team gather the information needed to address the issues and result in a better quality product.

Nevertheless, the performance metrics need evaluating before being used. An organization has to take into account the requirements of the customer, the quantifiable outputs of work, and the targets or goals of the projects in order to develop the appropriate performance metrics. Additionally, these metrics evolve depending on the state and the methodology an organization is following. For instance, before the Agile method, the defect tracking metric was one of the most relevant metrics, but once this new method was put forward, defect tracking was set aside as other metrics needed to be given more attention to. Furthermore, when an organization is still small, the metrics related to functional quality like customers' requirements and product design, among others, need to be regarded as paramount. But, as the project grows complex, the structural quality metrics, which are related to efficiency or maintenance of software, for example, need more considering.

Some important performance metrics that companies find useful are: defect density, which measures the amount of bugs that were tracked on a given period of time; customer problems, that need to be measured by the customers' response to the problems in terms of both product defects and troubles in usability; and defect removal effectiveness, which needs calculating depending on the amount of defects that were removed in contrast with the defects that remain in the software.





Let's Practice!

What is the importance of metrics to measure software quality?

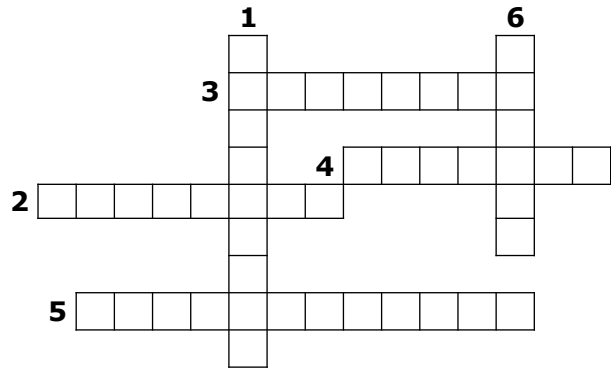
Instructions: Read the clues. Complete the crossword.

Across

- Performance metrics need to be used by software organizations for easily _____ problems.
- Defect tracking was _____ as other metrics needed to be given more attention to.
- The metrics help the team gather the information needed to _____ the issues and result in a better product.
- An organization has to take into account the _____ outputs of work, in order to develop the appropriate performance metrics.

Down

- Customer problems, need to be measured by the customers' response in terms of troubles in _____.
- _____ density is a performance metrics that companies find useful.

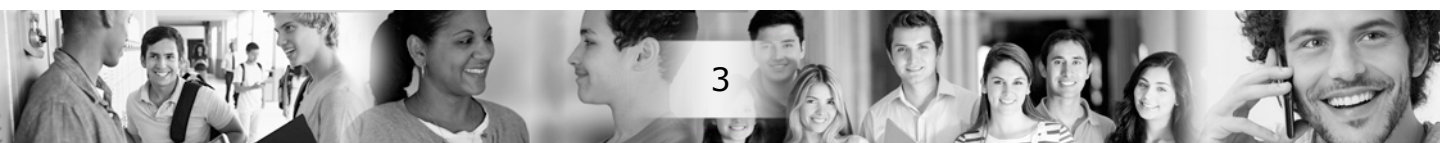


Let's Practice!

Remember that performance metrics are necessary to check the software quality.

Instructions: Complete the sentences by writing its corresponding word. Use capital letter where necessary. Use: need to be fixed / need evaluating / needs a lot of planning / need to be used / need to be fulfilled.

- Developing software can be an arduous work as it _____.
- Problems in performance metrics _____.
- Customers' requirements _____.
- Performance metrics _____ by software organizations.
- Performance metrics _____ before being used.





Extension Activity

What are performance metrics in real life?

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Look for information about some performance metrics in a company or in a product.
- Write a brochure describing the metrics and its relation with the product.
- Share the information with your partners.

Self-Assessment

Yes No

- | | | |
|--|-----------------------|-----------------------|
| 1. I read about performance metrics in the industry. | ☐ | ☐ |
| 2. I write about performance metrics. | ☐ | ☐ |
| 3. I talk about performance metrics in real life. | ☐ | ☐ |



Target Words

- **address** (verb): to deal with something, to regard something.
e.g. These metrics will help the manager and the team gather the information needed to address the issues and result in a better quality product.
- **quantifiable** (adjective): something that can be measured or given a certain quantity.
e.g. An organization has to take into account the requirements of the customer, the quantifiable outputs of work, and the targets or goals of the projects in order to develop the appropriate performance metrics.
- **defect** (noun): an anomaly in a product.
e.g. Customer problems, that need to be measured by the customers' response to the problems in terms of both product defects and troubles in usability.
- **usability** (noun): the extent to which something can be used.
e.g. Customer problems, that need to be measured by the customers' response to the problems in terms of both product defects and troubles in usability.
- **spot** (verb): to identify.
e.g. That's why performance metrics need to be used by software organizations for easily spotting problems in the development processes throughout the development of a product.
- **set aside** (phrasal verb): to reject or to put to one side.
e.g. Before the Agile method, the defect tracking metric was one of the most relevant metrics, but once this new method was put forward, defect tracking was set aside as other metrics needed to be given more attention to.



Learning Points

Some uses of the verb 'to need'

We can use the verb '**need**' with a passive infinitive or with an -ing verb when we want to express something that has to be done.

1. We can use the verb '**need**' with a **passive infinitive** when we want to express that a thing or a person has to receive an action.

Example: *The performance of a company **needs to be evaluated by** the manager.*

The structure we are going to use is the following:

Thing or person that receives the action + verb 'need' + to be + past participle of the main verb

Notice that we can either express the agent that is performing the action by using the preposition '**by**', or we can omit the agent that performs the action if the context permits it.

2. Another way of expressing that something has to be done or that something or someone has to receive an action, is by using the verb '**need**' with a gerund (**verb in -ing**). This structure will also have a passive sense.

Example: *The performance of the company **needs evaluating**.*

The structure we use is the following:

Thing or person that receives the action + verb 'need' + main verb in -ing

When we use a gerund to express a passive sense, we do not mention the agent that performed the action.

Examples

- *Developing software can be an arduous work as it **needs a lot of planning**, problems **need to be fixed**, and customers' requirements **need to be fulfilled**.*
- *Performance metrics **need evaluating before being used**.*
- *Before the Agile method, the defect tracking metric was one of the most relevant metrics, but once this new method was put forward, defect tracking was set aside as other metrics **needed to be given more attention to**.*





Learning Points

Understanding the Concept of Performance Metrics

Performance metrics determines or describes the performance of an organization regarding specific processes.

Choosing the right performance metrics is important for evaluating the performance of a software development project and avoiding inefficiency in the processes. If the metrics are not chosen correctly, the organization could be evaluating the performance of processes that are not that relevant for the overall project. Besides, when paying too much attention to process that are not very important, the truly crucial ones could be left out. This can happen especially if the organization decides to use mainly the performance metrics that are easier to measure.

It's important that an organization, whenever it assess the performance of a project, weights the effort to collect a metric against its value so that the collection of data does not become impractical.

There are seven criteria that an organization should take into account when choosing what the most relevant performance metrics are: safety, time, cost, resources, scope, quality, and actions. Nevertheless, the organization must evaluate what metrics are most important in accordance to how the organization works and the nature of the project it is working on.

Finally, when choosing performance metrics, there are three important steps that can help discriminate what the most relevant measurements are:

1. Establish critical requirements.
2. Identify specific and quantifiable outputs of work (measurable results which can be used to determine the quality and performance of the product).
3. Establish targets that help measure results.

